Theorem (Functional Calculus).

Let A be a unital C^*-algebra and $a \in A$ normal $\left(a^*a = aa^*\right)$.

Then there is a unique isometric $*$-homomorphism

$$C(\sigma(a)) \longrightarrow A$$

such that

$$\mathrm{id}_{\sigma(a)} \longmapsto a, \qquad 1_{\sigma(a)} \longmapsto 1_A.$$

Ex.

$$\mathrm{id}_{\sigma(a)} : \sigma(a) \to \mathbb{C}, \qquad \mathrm{id}_{\sigma(a)}(z) = z,$$
$$1_{\sigma(a)} : \sigma(a) \to \mathbb{C}, \qquad 1_{\sigma(a)}(z) = 1.$$

Base result: Section 2 notes, Theorem 2.22 (Functional Calculus).

Original typed annotation in the photo: "by means of isometric it means that the norm is always one."

Correction. "Isometric" means norm preserving:

$$\|F(f)\|_A = \|f\|_\infty \quad \text{for every } f,$$

not that every norm is equal to 1.

Proof.

Put

$$B := C^*(a, 1_A).$$

Since a is normal, B is a unital commutative C^*-algebra.

Tarea 1, Problem 3(d): a $*$-commuting generating family generates a commutative involutive Banach algebra; see also the main study notes, §4.3.

By the Gelfand–Naimark theorem,

$$\Lambda_B : B \longrightarrow C(\Omega(B)), \qquad b \longmapsto \widehat{b},$$

is an isometric $*$-isomorphism.

Section 2 notes, Theorem 2.20 (commutative Gelfand–Naimark).

We already showed

$$\Omega(B) = \Omega(C^*(a, 1_A)) \cong \sigma(a)$$

via the homeomorphism

$$\Phi : \Omega(B) \longrightarrow \sigma(a), \qquad \Phi(\varphi) = \varphi(a).$$

Section 2 notes, Proposition 2.18.

Then

$$\Phi_* : C(\Omega(B)) \longrightarrow C(\sigma(a)), \qquad f \longmapsto f \circ \Phi^{-1},$$

is an isometric $*$-isomorphism.

Indeed,

$$\begin{aligned} \|\Phi_*(f)\|_\infty &= \|f \circ \Phi^{-1}\|_\infty \\ &= \sup_{\lambda \in \sigma(a)} \left| f\left(\Phi^{-1}(\lambda)\right) \right| \\ &= \sup_{\varphi \in \Omega(B)} |f(\varphi)| \\ &= \|f\|_\infty. \end{aligned}$$

The spectral description used here is

$$\sigma(a) = \{\lambda \in \mathbb{C} : \lambda 1_A - a \notin GL(A)\} = \{\varphi(a) : \varphi \in \Omega(B)\},$$

and, if $\lambda = \varphi(a)$, then

$$\Phi^{-1}(\lambda) = \Phi^{-1}(\varphi(a)) = \varphi.$$

Definition of spectrum: Part 1, Definition 1.11. Spectrum via characters: Part 1, Theorem 1.22, applied inside the commutative algebra B; spectral permanence is Section 2, Proposition 2.17.

Surjectivity of Φ_*.

Let $f \in C(\sigma(a))$. Since Φ is a homeomorphism, define

$$g := f \circ \Phi \in C(\Omega(B)).$$

Then

$$\Phi_*(g) = g \circ \Phi^{-1} = f \circ \Phi \circ \Phi^{-1} = f,$$

so Φ_* is surjective.

Let's consider

$$\Phi_* \circ \Lambda_B : B \longrightarrow C(\sigma(a)).$$

Let $\lambda \in \sigma(a)$ and put

$$\varphi := \Phi^{-1}(\lambda) \in \Omega(B).$$

For the unit,

$$\begin{aligned}\big((\Phi_* \circ \Lambda_B)(1_A)\big)(\lambda) &= \Phi_*\big(\Lambda_B(1_A)\big)(\lambda) \\ &= \widehat{1_A}\big(\Phi^{-1}(\lambda)\big) \\ &= \widehat{1_A}(\varphi) \\ &= \varphi(1_A) \\ &= 1 \\ &= 1_{\sigma(a)}(\lambda).\end{aligned}$$

For a,

$$\begin{aligned}\big((\Phi_* \circ \Lambda_B)(a)\big)(\lambda) &= \Phi_*\big(\Lambda_B(a)\big)(\lambda) \\ &= \widehat{a}\big(\Phi^{-1}(\lambda)\big) \\ &= \widehat{a}(\varphi) \\ &= \varphi(a) \\ &= \lambda \\ &= \mathrm{id}_{\sigma(a)}(\lambda).\end{aligned}$$

Instructor's annotation in the photograph:

"falta demostrar que es un isomorfismo isométrico".

Completion of that missing verification. Since Λ_B is an isometric $*$-isomorphism by Theorem 2.20 and Φ_* is an isometric $*$-isomorphism by the calculation above, their composition is also an isometric $*$-isomorphism:

$$\Phi_* \circ \Lambda_B : B \xrightarrow{\cong} C(\sigma(a)).$$

Moreover, $\Phi_* \circ \Lambda_B$ is an isometric $*$-isomorphism.

Therefore

$$(\Phi_* \circ \Lambda_B)^{-1} : C(\sigma(a)) \longrightarrow B \subseteq A$$

is an isometric $*$-isomorphism, and

$$(\Phi_* \circ \Lambda_B)^{-1}\big(\mathrm{id}_{\sigma(a)}\big) = a, \qquad (\Phi_* \circ \Lambda_B)^{-1}\big(1_{\sigma(a)}\big) = 1_A.$$

This is exactly the construction used in the proof of Section 2, Theorem 2.22.

Source-result key used in this transcription.

- *[P1] Jorge Castillejos, C^*-álgebras: Teoría Espectral en álgebras de Banach: Definition 1.11, Theorem 1.22, Definition 1.27.*
- *[C2] Jorge Castillejos, C^*-álgebras, Section 2: Proposition 2.18, Theorem 2.19, Theorem 2.20, Theorem 2.22.*
- *[T1] C^*-álgebras, Tarea 1: Problem 3(d), and Problem 9 for a spectrum-preserving commutative reduction.*
- *[N] Enrique Díaz Ocampo, Notas de estudio: teoría espectral en álgebras de Banach y C^*-álgebras, §3.4.*